

REQUIREMENT ANALYSIS

Electric Vehicle Analytics Dashboard using Flask & Tableau E-CarStart

Project Information

Project	Electric Vehicle Analytics Dashboard
Frontend	HTML, CSS, Bootstrap
Backend	Python Flask
Visualization	Tableau Public
Dataset	Electric Vehicle Registration Dataset

Introduction

This document captures the functional and non-functional requirements for the Electric Vehicle Analytics Dashboard web application.

Requirement Gathering

Requirements were identified based on user needs, project objectives, and analysis of EV registration data.

Functional Requirements

- View Home page
- Access Dashboard
- Access Story
- Navigate website
- Display embedded Tableau visualizations
- Responsive user interface

Non-Functional Requirements

Performance, usability, availability, maintainability, reliability, security, and scalability.

Software Requirements

Python 3.x, Flask 3.x, VS Code, Tableau Public, Git, Bootstrap 5.

Hardware Requirements

Processor: Intel i3 or above

RAM: 4 GB minimum

Storage: 500 MB

Internet connection required.

Technology Stack

Python, Flask, HTML5, CSS3, Bootstrap, Tableau Public.

User Requirements

Students, Researchers, Government Agencies, Policy Makers, Business Analysts, General Public.

Use Case Summary

Users can navigate the website, explore the dashboard, and interact with the Tableau story to analyze EV trends.

Conclusion

The identified requirements provide a clear foundation for developing a reliable, interactive EV Analytics Dashboard.